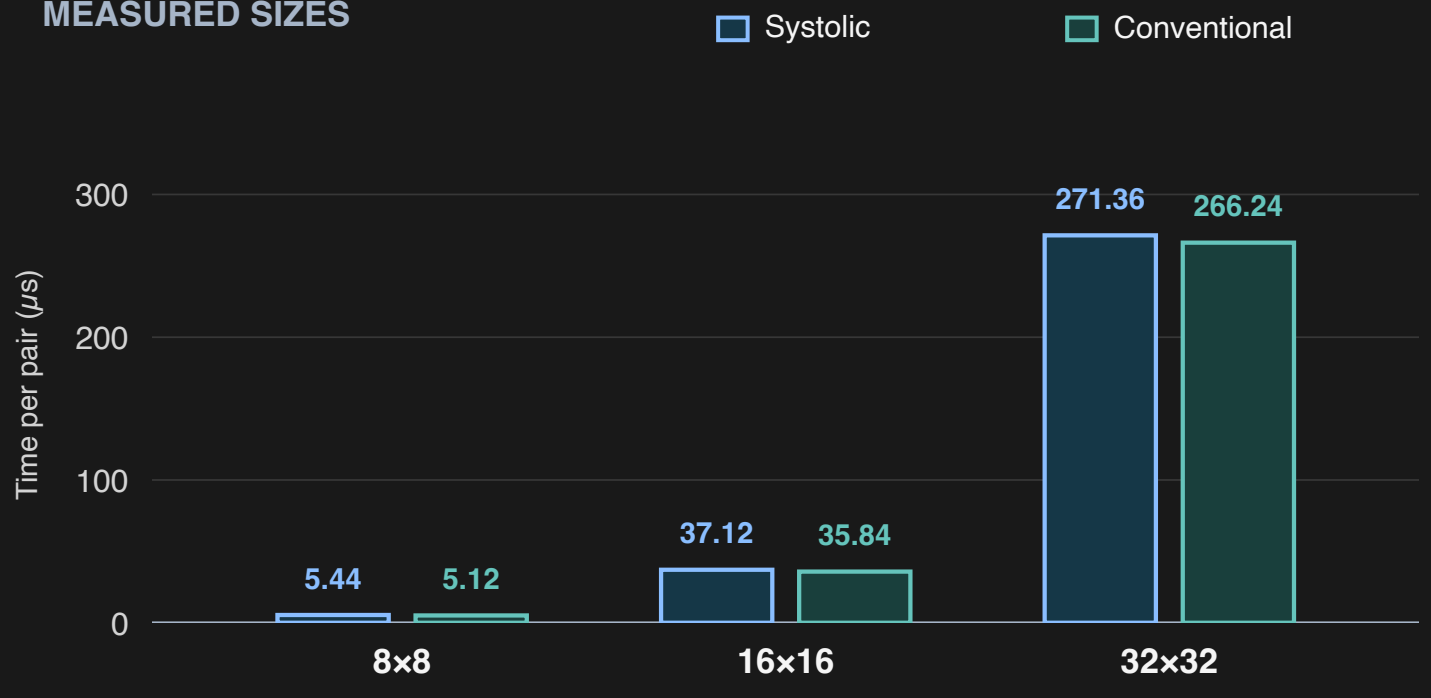
COMPUTE TIME

Systolic array vs. conventional matrix multiplier

4 MACs in each core | **100 MHz** | **SAED32 RVT · 1.05 V · 25 °C**

MEASURED SIZES



At 32×32: conventional uses 1.9% less time.

64×64 PROJECTION

Four MACs; larger matrices

SYSTOLIC

2,068.48 µs

CONVENTIONAL

2,048.00 µs

Schedule formula at 100 MHz

Not simulated or synthesized

Accepted start through done, including internal tile loading; external input load/readback excluded.

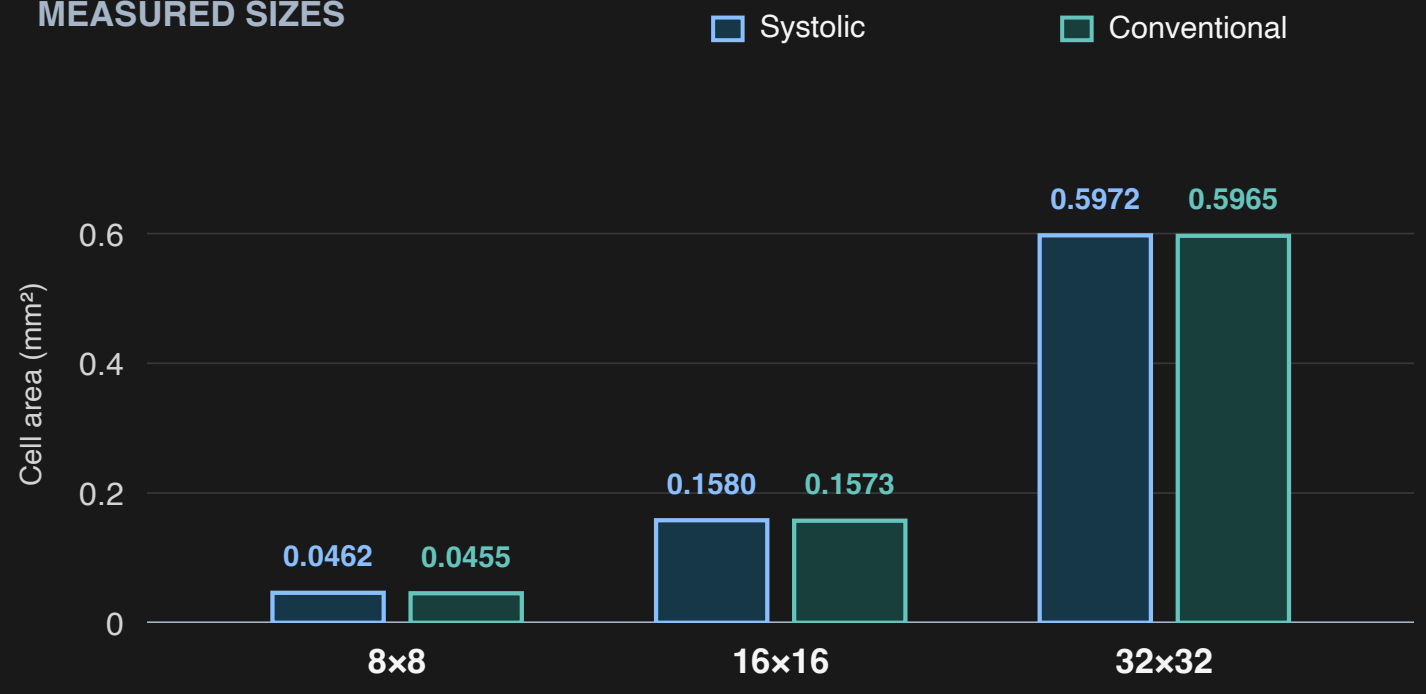
64×64 assumes extended size support, the same schedule, and successful timing closure.

MAPPED CELL AREA

Systolic array vs. conventional matrix multiplier

4 MACs in each core | **100 MHz** | **SAED32 RVT · 1.05 V · 25 °C**

MEASURED SIZES



Nearly tied at 32×32: 0.119% area gap.

64×64 PROJECTION

Four MACs; larger matrices

SYSTOLIC

2.3590 mm²

Model range: 2.11-2.36 mm²

CONVENTIONAL

2.3582 mm²

Model range: 2.13-2.36 mm²

Main estimate: $aN^2 + b$ fit

Not simulated or synthesized

Mapped standard-cell area, including register-based storage; placement and routing excluded.

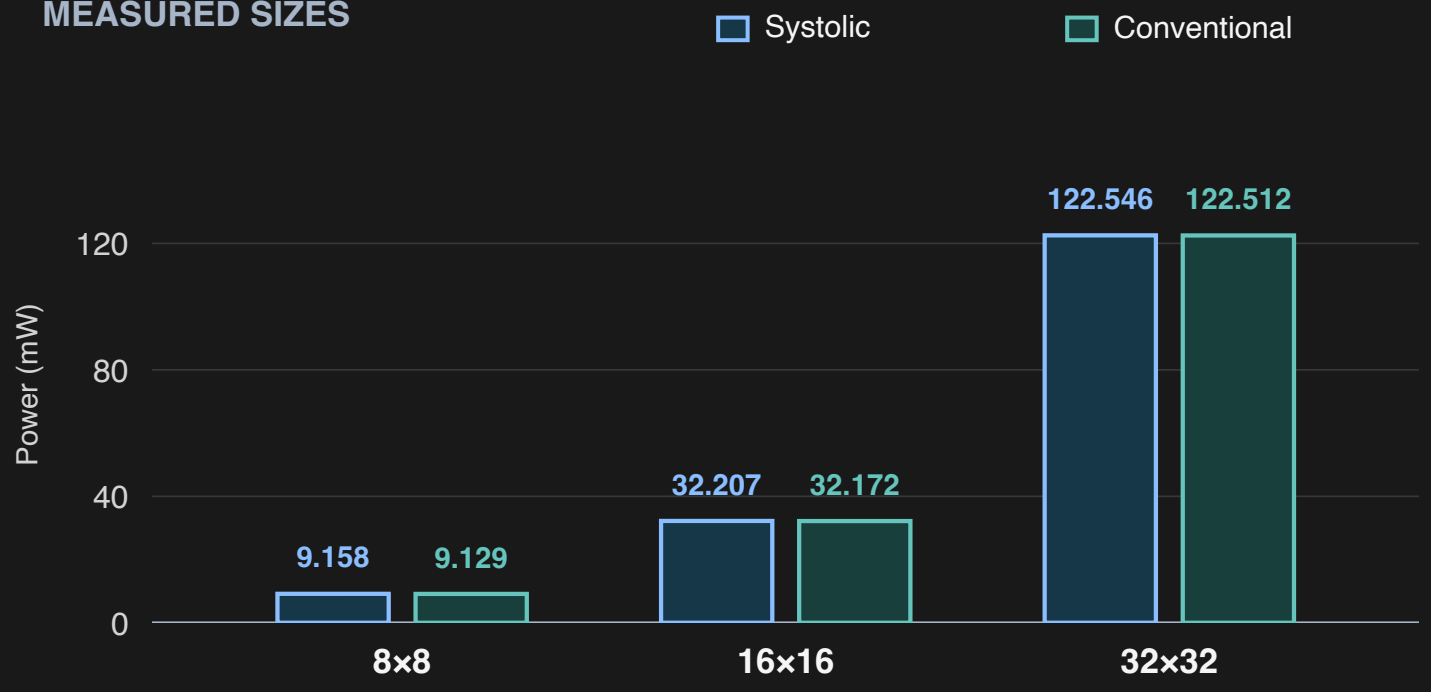
Prediction ranges compare three models; they are not confidence intervals.

AVERAGE TOTAL POWER

Systolic array vs. conventional matrix multiplier

4 MACs in each core | **100 MHz** | **SAED32 RVT · 1.05 V · 25 °C**

MEASURED SIZES



Nearly tied at 32×32: 0.028% power gap.

64×64 PROJECTION

Four MACs; larger matrices

SYSTOLIC

485.009 mW

Model range: 442.45-485.01 mW

CONVENTIONAL

484.960 mW

Model range: 443.04-484.96 mW

Main estimate: $aN^2 + b$ fit

Not simulated or synthesized

Dynamic + leakage over load/compute/readback. Mean of 3 seeds × 100 random pairs.

Bar whiskers: seed min-max (very small). Prediction ranges: model choice, not confidence intervals.